$$= \frac{\sin 55^\circ}{\sin 55^\circ} + \frac{\cos 78^\circ}{\cos 78^\circ} - \frac{\sin 72^\circ}{\sin 72^\circ}$$

$$= 1 + 1 - 1 = 1$$

Example 6.10 (i) If $\operatorname{cosec} A = \sec 34^\circ$, then find A (ii) If $\tan B = \cot 47^\circ$, then find B.

Solution

(i) We know that $\operatorname{cosec} A = \sec(90^\circ - A)$

$\sec(90^\circ - A) = \sec(34^\circ)$

$90^\circ - A = 34^\circ$

We get $A = 90^\circ - 34^\circ$

$A = 56^\circ$

(ii) We know that $\tan B = \cot(90^\circ - B)$

$\cot(90^\circ - B) = \cot 47^\circ$

$90^\circ - B = 47^\circ$

We get $B = 90^\circ - 47^\circ$

$B = 43^\circ$

Exercise 6.3

Find the value of the following:

(i) $\left(\frac{\cos 47^\circ}{\sin 43^\circ}\right)^2 + \left(\frac{\sin 72^\circ}{\cos 18^\circ}\right)^2 - 2\cos^2 45^\circ$ (ii) $\frac{\cos 70^\circ}{\sin 20^\circ} + \frac{\cos 59^\circ}{\sin 31^\circ} + \frac{\cos\theta}{\sin(90^\circ - \theta)} - 8\cos^2 60^\circ$

(iii) $\tan 15^\circ \tan 30^\circ \tan 45^\circ \tan 60^\circ \tan 75^\circ$

(iv) $\frac{\cot\theta}{\tan(90^\circ - \theta)} + \frac{\cos(90^\circ - \theta)\tan\theta\sec(90^\circ - \theta)}{\sin(90^\circ - \theta)\cot(90^\circ - \theta)\operatorname{cosec}(90^\circ - \theta)}$

Thinking Corner

(i) What are the minimum and maximum values of $\sin\theta$?

(ii) What are the minimum and maximum values of $\cos\theta$?

6.4 Method of using Trigonometric Table

We have learnt to calculate the trigonometric ratios for angles 0°, 30°, 45°, 60° and 90°. But during certain situations we need to calculate the trigonometric ratios of all the other acute angles. Hence we need to know the method of using trigonometric tables.

One degree (1°) is divided into 60 minutes ($60'$) and one minute ($1'$) is divided into 60 seconds ($60''$). Thus, $1^\circ = 60'$ and $1' = 60''$.

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